Comparative Evaluation of Traditional Machine Learning and Recurrent Neural Network Models for Multi-Class News Headline Classification

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***Abstract*—This paper presents a comparative study of traditional machine learning and recurrent neural network models for multi-class news headline classification. The task is to classify news headlines into four categories: Business, Science and Technology, Sports, and World News. Three preprocessing settings were evaluated: raw text, extreme preprocessing, and optimum preprocessing. Two text representation methods were used: TF-IDF and Skip-gram Word2Vec embeddings. Logistic Regression and a Deep Neural Network were trained using TF-IDF features, while SimpleRNN, GRU, LSTM, BiSimpleRNN, BiGRU, and BiLSTM models were trained using Skip-gram embeddings. The best result was achieved by the GRU model using Skip-gram embeddings with optimum preprocessing, obtaining a test accuracy of 91.89% and a macro-F1 score of 91.88%. The results show that gated recurrent models are effective for learning sequential patterns from headlines, while TF-IDF remains a strong baseline for short-text classification.**

***Index Terms*—News classification, text classification, natural language processing, TF-IDF, Skip-gram, recurrent neural networks.**

I. Introduction

News articles are produced in large volumes across many domains, making automatic news categorization an important natural language processing task. Manual classification is time-consuming and difficult to scale, especially when headlines need to be organized for search, recommendation, filtering, or topic-based browsing. Text classification can automate this process by learning patterns from labeled examples and assigning headlines to appropriate categories.

This project compares traditional feature-based methods and neural sequence-based models for classifying news headlines. Since headlines are short and often contain domain-specific keywords, both TF-IDF and Skip-gram Word2Vec representations were used. TF-IDF captures the importance of words across documents, while Word2Vec provides dense vector representations that capture semantic relationships between words.

The dataset contains news headlines labeled into four categories: Business, Science and Technology, Sports, and World News. Three preprocessing versions were created to compare the effect of cleaning level on classification performance: raw, extreme, and optimum. Logistic Regression and a Deep Neural Network were trained using TF-IDF features, while six recurrent neural models were trained using Skip-gram embeddings. The goal was to identify the best combination of preprocessing, representation, and model architecture for multi-class news headline classification.

II. Methodology

A. Dataset and Exploratory Data Analysis

The dataset used in this study consists of one training file and one testing file. Each record contains a news headline and its corresponding topic label. The classification task includes four categories: Business, Science and Technology, Sports, and World News. Table I summarizes the dataset.

TABLE I
Dataset Summary

Item	Value
Training records	104,763
Test records	12,000
Input column	News Headline
Target column	News Topic
Number of classes	4
Classes	Business, Sci/Tech, Sports, World

Before model training, exploratory data analysis was performed to examine missing values, class distribution, headline length, and category-specific vocabulary. The class distribution showed that the dataset was slightly imbalanced, with Science and Technology and Sports containing more samples than Business and World News. This motivated the use of macro-F1 score in addition to accuracy. The class distribution and headline length distribution are shown in Fig. 1 and Fig. 2.

Word cloud visualizations were generated for each category to observe frequent and topic-specific terms. These word clouds helped identify broad vocabulary patterns across the four classes. For example, Sports headlines contained words related to matches, teams, and players, while Business headlines included terms related to companies, markets, and

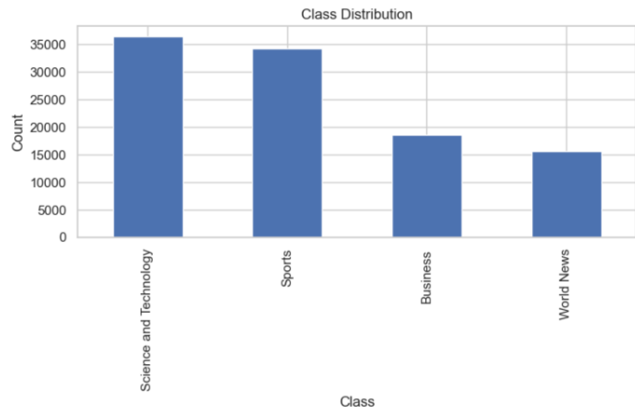

Fig. 1. Class distribution of the four news categories.

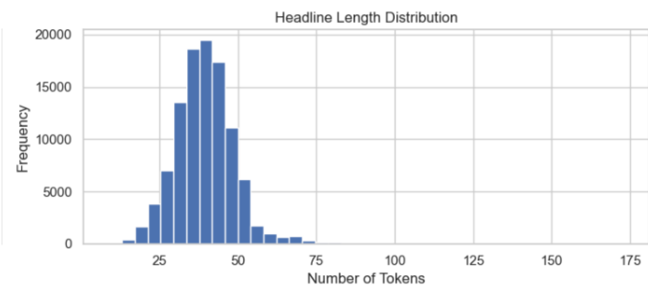

Fig. 2. Distribution of headline lengths in the training dataset.

financial activities. The category-wise word clouds are shown in Fig. 3.

Fig. 3. Category-wise word cloud visualizations for Business, Science and Technology, Sports, and World News.

In addition to word clouds, top unigram analysis was performed to provide a more quantitative view of frequent terms in each category. The unigram plots in Fig. 4 show clear lexical differences across the classes, supporting the use of both TF-IDF and word embedding representations.

After EDA, the original training dataset was split into training and validation sets using stratified splitting. This preserved the class proportions across the subsets. From the 104,763

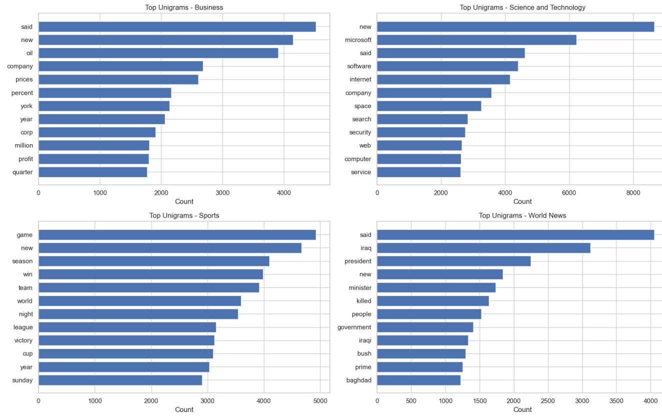

Fig. 4. Top unigram terms for each news category.

training records, 89,048 samples were used for training and 15,715 samples were used for validation. The separate 12,000-sample test set was kept unchanged and used only for final evaluation.

B. Data Preprocessing and Text Representation

Text preprocessing was performed because the raw news headlines contained HTML tags, newline characters, punctuation, inconsistent casing, and formatting noise. Since machine learning and deep learning models require numerical inputs, the headlines were converted into cleaned textual formats before feature extraction.

Three preprocessing versions were created. The **raw version** kept the original headline text unchanged and was used as a baseline. The **extreme** preprocessing version removed HTML tags, converted text to lowercase, kept only alphabetic tokens, removed stopwords, and applied Porter stemming. This version reduced noise but also removed some information that could be useful for classification. The **optimum** preprocessing version applied balanced cleaning by removing HTML tags, converting text to lowercase, cleaning punctuation, and removing extra spaces. Unlike the extreme version, stopwords and numbers were preserved because they may carry useful information in short news headlines.

After preprocessing, the categorical class labels were converted into numerical labels using label encoding. Two text representation techniques were then used: TF-IDF and Skip-gram Word2Vec. For TF-IDF, headlines were converted into sparse vectors based on term importance. Unigram and bigram features were used to capture both individual words and short word combinations. TF-IDF features were used with Logistic Regression and the Deep Neural Network.

For Skip-gram Word2Vec, dense word embeddings were trained from the training headlines. Each headline was tokenized, converted into a sequence of word indices, and padded to a fixed length. These sequence representations were used as input for recurrent neural network models. The Word2Vec embedding matrix was used to initialize the embedding layer of the sequence-based models.

C. Model Architectures

This study evaluated both traditional machine learning and neural network models for multi-class news headline classification. Logistic Regression was used as the main traditional baseline with TF-IDF features because it is efficient and effective for sparse high-dimensional text representations. A TF-IDF-based Deep Neural Network was also trained to determine whether non-linear dense layers could improve performance over the linear baseline. The network used fully connected layers with ReLU activation, batch normalization, and dropout regularization, followed by an output layer with four neurons corresponding to the four news categories.

For sequence-based learning, Skip-gram Word2Vec embeddings were used with six recurrent neural architectures: SimpleRNN, GRU, LSTM, BiSimpleRNN, BiGRU, and BiLSTM. The embedding layer was initialized using the Word2Vec embedding matrix trained on the training headlines. SimpleRNN was included as a basic recurrent baseline, while GRU and LSTM were used because their gating mechanisms help control information flow and reduce the vanishing gradient problem. Bidirectional versions of these models were also evaluated to determine whether using both forward and backward context improves classification performance.

The GRU architecture produced the best result in this study. It first converted each headline into a padded sequence of word indices. These indices were passed through the Word2Vec embedding layer, followed by a GRU encoder that learned sequential dependencies between tokens. The final hidden representation was then passed through dropout and fully connected layers before the final output layer. This design allowed the model to capture meaningful word-order patterns while maintaining lower computational complexity than LSTM-based models.

All neural models were trained using cross-entropy loss and the Adam optimizer. Batch size was set to 256 for the TF-IDF Deep Neural Network and 128 for the Skip-gram-based recurrent models. Early stopping was applied based on validation performance to reduce overfitting. Model performance was evaluated using validation accuracy, validation macro-F1, test accuracy, test macro-F1, classification reports, and confusion matrices.

D. Hyperparameter Selection

The hyperparameters were selected based on dataset characteristics, preprocessing type, computational efficiency, and validation performance. Since raw text retained more tokens and noise, it used the largest maximum sequence length. Extreme preprocessing removed stopwords and applied stemming, so it used the shortest sequence length. The optimum version used a balanced sequence length because it removed formatting noise while preserving useful words and numbers.

For TF-IDF, the maximum number of features was adjusted according to the vocabulary size produced by each preprocessing strategy. Similarly, Word2Vec embedding dimensions were varied so that the optimum version could preserve richer semantic information. Table II summarizes the main hyperparameter settings. The same training objective, optimizer, and evaluation metrics were used across neural models to ensure a fair comparison.

TABLE II
MAIN HYPERPARAMETER SETTINGS

Setting	Raw	Extreme	Optimum
TF-IDF max features	25,000	18,000	22,000
Word2Vec vector size	120	100	140
Maximum sequence length	60	40	50
TF-IDF DNN batch size	256	256	256
RNN batch size	128	128	128

III. RESULTS AND DISCUSSION

The performance of the models was evaluated using accuracy and macro-F1 score. Accuracy measures the percentage of correctly classified samples, while macro-F1 gives equal importance to all four classes by averaging the F1-score of each class. Since the dataset is slightly imbalanced, macro-F1 is especially useful for comparing model performance fairly across Business, Science and Technology, Sports, and World News.

The results in Table III show that the highest-performing model was the GRU model using Skip-gram Word2Vec with optimum preprocessing, achieving a test accuracy of 91.89% and a test macro-F1 score of 91.88%. This indicates that gated recurrent architectures are highly effective for learning sequential patterns from news headlines. The next strongest results were obtained by LSTM and bidirectional recurrent models, which also achieved test macro-F1 scores above 91.5% in several preprocessing settings.

TF-IDF-based models also performed competitively. The TF-IDF Deep Neural Network with extreme preprocessing achieved 91.48% test accuracy and 91.47% test macro-F1, while Logistic Regression with extreme preprocessing achieved 91.29% test accuracy and 91.27% test macro-F1. These results show that TF-IDF remains a strong representation for short-text classification, although the best individual result was obtained by the Skip-gram-based GRU model.

The weakest model was the SimpleRNN trained on raw text, which achieved 83.30% test accuracy and 83.36% test macro-F1. This result shows that basic recurrent networks are less effective when applied to noisy raw text. In contrast, GRU and LSTM models performed more consistently because their gating mechanisms helped retain useful sequential information.

Fig. 5 shows the test macro-F1 scores across model and preprocessing combinations. The gated recurrent models, especially GRU, LSTM, BiGRU, and BiLSTM, achieved consistently strong performance compared to SimpleRNN-based models. The best score was obtained by the GRU model with optimum preprocessing, confirming that gated recurrent

TABLE III
COMPARATIVE PERFORMANCE OF ALL MODELS

Rep.	Model	Prep.	Val F1	Test F1	Test Acc.
Skip-gram	GRU	Opt.	93.19	91.88	91.89
Skip-gram	LSTM	Ext.	92.52	91.73	91.75
Skip-gram	LSTM	Opt.	92.82	91.69	91.73
Skip-gram	BiLSTM	Opt.	92.57	91.68	91.69
Skip-gram	BiGRU	Raw	92.98	91.67	91.69
Skip-gram	BiGRU	Opt.	93.45	91.67	91.68
Skip-gram	BiGRU	Ext.	93.06	91.53	91.55
Skip-gram	BiLSTM	Raw	92.75	91.51	91.50
Skip-gram	BiLSTM	Ext.	93.01	91.50	91.51
Skip-gram	GRU	Raw	92.15	91.20	91.18
Skip-gram	GRU	Ext.	92.82	91.00	91.03
Skip-gram	LSTM	Raw	90.98	90.22	90.26
Skip-gram	SimpleRNN	Ext.	89.24	89.42	89.45
Skip-gram	BiSimpleRNN	Opt.	89.35	89.34	89.35
Skip-gram	BiSimpleRNN	Ext.	90.15	89.24	89.30
Skip-gram	BiSimpleRNN	Raw	89.00	86.77	86.85
Skip-gram	SimpleRNN	Opt.	86.22	86.03	86.13
Skip-gram	SimpleRNN	Raw	82.26	83.36	83.30
TF-IDF	Deep NN	Ext.	93.37	91.47	91.48
TF-IDF	Logistic Reg.	Ext.	92.84	91.27	91.29
TF-IDF	Logistic Reg.	Opt.	92.76	91.18	91.21
TF-IDF	Deep NN	Opt.	92.82	90.68	90.70
TF-IDF	Deep NN	Raw	93.19	89.83	89.81
TF-IDF	Logistic Reg.	Raw	93.11	88.87	88.83

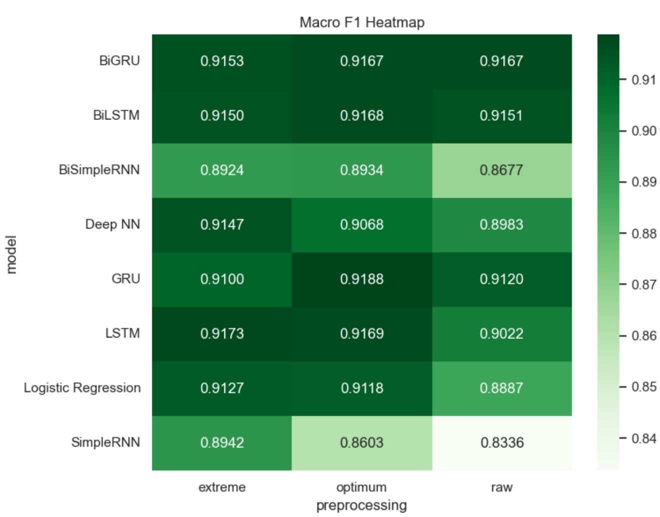

Fig. 5. Test macro-F1 heatmap across model and preprocessing combinations.

architectures are more effective for learning headline sequence patterns.

Table IV shows the average macro-F1 score across all models for each preprocessing strategy. Extreme preprocessing achieved the highest average macro-F1 score, followed by optimum preprocessing. However, the best individual model was still obtained using optimum preprocessing. This suggests that aggressive preprocessing can improve average performance, while balanced preprocessing may better preserve useful information for the strongest sequence model.

TABLE IV
AVERAGE TEST MACRO-F1 BY PREPROCESSING METHOD

Preprocessing	Average Test Macro-F1 (%)
Extreme	**90.90**
Optimum	90.52
Raw	89.18

TABLE V
AVERAGE TEST MACRO-F1 BY MODEL

Representation	Model	Avg. Test Macro-F1 (%)
Skip-gram	BiGRU	**91.62**
Skip-gram	BiLSTM	91.56
Skip-gram	GRU	91.36
Skip-gram	LSTM	91.21
TF-IDF	Deep NN	90.66
TF-IDF	Logistic Regression	90.44
Skip-gram	BiSimpleRNN	88.45
Skip-gram	SimpleRNN	86.27

Table V shows that BiGRU achieved the highest average macro-F1 score across preprocessing versions, followed closely by BiLSTM, GRU, and LSTM. Although the best single run was obtained by GRU with optimum preprocessing, the average results indicate that bidirectional gated recurrent models were generally reliable across different preprocessing settings.

Fig. 6 shows the test macro-F1 scores across all experimental combinations. The GRU model using Skip-gram embeddings with optimum preprocessing achieved the highest macro-F1 score. Most gated recurrent models, including GRU, LSTM, BiGRU, and BiLSTM, performed strongly, while SimpleRNN-based models produced lower scores. The TF-IDF-based models also remained competitive, especially under extreme preprocessing, confirming that TF-IDF is a strong baseline for short-text classification.

The classification reports were examined to compare class-wise precision, recall, and F1-score. The best model maintained strong performance across all four classes, showing that the improvement was not limited to the majority categories. This is important because the dataset was slightly imbalanced. In contrast, weaker SimpleRNN runs showed lower and less stable class-wise scores, especially when trained on raw text. This supports the use of macro-F1 as a main evaluation metric.

Fig. 7 compares the confusion matrices of the best and worst runs. The best run, GRU with optimum preprocessing and Skip-gram embeddings, shows a strong diagonal pattern, meaning that most samples were correctly classified across all four classes. In contrast, the worst run, SimpleRNN with raw text, produced more off-diagonal errors, especially between semantically closer categories. This confirms that both preprocessing and gated recurrent architectures improved class-wise

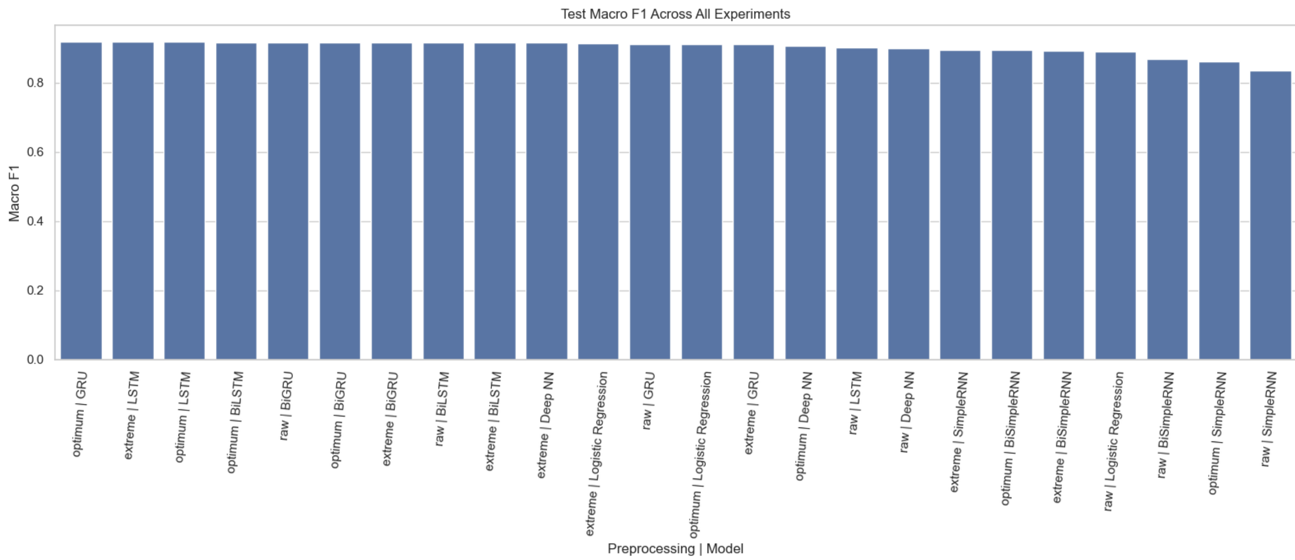

Fig. 6. Test macro-F1 scores across all preprocessing, representation, and model combinations.

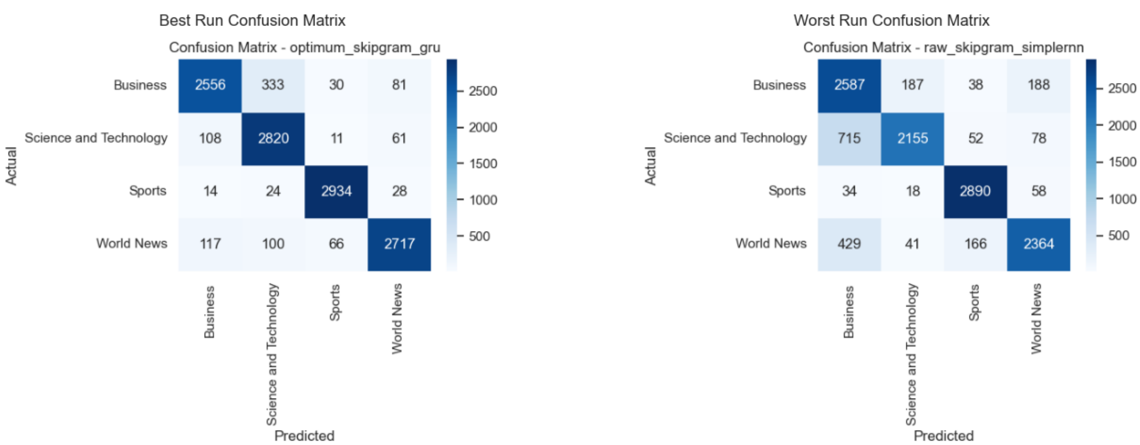

Fig. 7. Confusion matrices of the best-performing and worst-performing runs.

prediction performance.

Overall, the results show that both traditional machine learning and recurrent neural network models are effective for news headline classification. TF-IDF-based models provided strong and efficient baselines, while Skip-gram-based gated recurrent models achieved the strongest individual and average sequence-model results. Among all tested combinations, GRU with Skip-gram Word2Vec and optimum preprocessing was selected as the best model because it achieved the highest test accuracy and test macro-F1 score.

IV. Conclusion

This project presented a comparative study of traditional machine learning and recurrent neural network models for multi-class news headline classification. Three preprocessing strategies were evaluated with two representation methods: TF-IDF and Skip-gram Word2Vec. The results showed that preprocessing improved performance over raw text, with extreme preprocessing producing the highest average macro-F1 and optimum preprocessing producing the best individual model result.

Among all individual experiments, the GRU model using Skip-gram Word2Vec with optimum preprocessing achieved the best result, with 91.89% test accuracy and 91.88% test macro-F1. TF-IDF-based Logistic Regression and Deep Neural Network models also remained highly competitive, showing that traditional feature-based methods are strong baselines for short-text classification. Overall, the findings indicate that gated recurrent models are effective for learning sequential patterns from headlines, while TF-IDF remains efficient and reliable for multi-class news classification.

One limitation of this study is that the models were trained only on headline text, which is short and may not always contain enough context to distinguish between similar topics. In addition, the Word2Vec embeddings were trained only on the project dataset rather than using a larger external corpus. Future improvements could include using transformer-based models such as BERT, incorporating full article text, and performing broader hyperparameter tuning to further improve classification performance.

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